

# Mihails Bazilovs

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## EDUCATION

### HAN University of Applied Sciences

BSc Mechanical engineering (8.0 AVG)

Arnhem

Sep. 2024 – July 2028

### Relevant coursework

Dynamics, Energy systems, Fluid mechanics, Material science, Statics, Structural mechanics, Thermodynamics

## EXPERIENCE

### CAD Constructor/Design Engineer

JSC "Atri"

Feb. 2026 – Present

Arnhem

- Designing the fuselage and internal structure for a high-velocity drone
- Designing the cooling system for electrical components
- Manufacturing and testing the prototype

## PROJECTS

### Battle Bot Weapon System Engineer | CAD, Structural Analysis, MS Excel

Feb. 2026 – June 2026

- Using Microsoft Excel created a spreadsheet which was used in order to design the weapon's geometry and select materials with respect to the key failure modes
- Designed full CAD model using SolidWorks
- Created production drawings for further manufacturing

### GE90 turbofan engine case study | Flow analysis, CAD, Control systems, Heat transfer

Apr. 2026 – June 2026

- Using MS Excel calculated heat transfer and temperatures after combustion within the GE90 turbofan engine and validated results using SolidWorks Flow Analysis
- Derived first order differential equation for the heat transfer and using MATLAB Simulink created a control system to adjust fuel contents in the air-fuel mixture

### Redesign of a vibratory conveyor | FEA, Frequency analysis, Bending calculations

Sep. 2025 – Jan 2026

- Identified excessive deformation in the vibratory conveyor's frame by performing hand calculations and verified them by simulating deformation in Solidworks
- Reduced the maximum bending by approximately 3.5 times using FEA
- Determined necessary springs' stiffness and specifications for the eccentric motor so the vibratory conveyor fits client's requirements and validated selection using SolidWorks vibrations analysis

### Automated Guided Vehicle (AGV) | Team management, Mechanical calculations, CAD

Sep. 2025 – Jan 2026

- Led the interdisciplinary team of 8 people by planning and allocating tasks and running weekly meetings
- Created a detailed CAD model to ensure correct interpretation of the design by the client
- Reduced costs of the project by developing a gearbox for the available low-torque motors

### Mechanical battery | Energy calculation, CAD, Material selection, Manufacturing

Apr. 2025 – June 2025

- Carried out in-depth field research on a flywheel and developed a mechanical clutch mechanism that reduced design complexity and made it more cost efficient
- Redesigned the flywheel in a very short time due to unforeseen circumstances, while maintaining efficiency

### Wind-powered water pump system | Stress analysis, CAD, Metalworking

Sep. 2024 – Jan. 2025

- Optimized windmill blades by researching existing solutions and performing flow simulation in SolidWorks
- Manufactured windmill blades and the windmill hub using lathe and tools for metal sheet shaping

## SKILLS

**Technical skills:** SolidWorks (intermediate), MATLAB (beginner), MS Office (intermediate), GitLab, Metalworking, Welding, 3D printing, FEA (beginner), Codesys (beginner)

**Professional skills:** Rapid CAD modeling, Prototyping, Technical report writing, Public presentation skills

**Certifications:** VOL - VCA, MATLAB Onramp

**Languages:** Latvian (native), English (fluent), Russian (native), German (basic), Dutch (basic)

**Hobbies:** PC building, Rock climbing, Guitar